Reasoning State Lives in the Chain of Thought, Not in J-Space: Written Values Are Addressable Memory

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Abstract

If chain-of-thought text reports computation carried out in a model's internal concept workspace, then ablating that workspace should impair reasoning and rewriting the text should not matter. We find the reverse on both counts. Ablating the concepts a fitted Jacobian lens (J-space) reads from the residual stream, at the strongest dose that leaves ordinary generation intact, changes nothing about multi-step reasoning; rewriting what sits at a written value's token controls the answer. The intermediate values a model writes in its chain of thought are the memory the computation runs on: each written value is held at its own token position as state that later computation reads back through attention, and this stored value, and never the internal workspace, determines the final answer. Overwriting the internal state at a written value's token with the state a different number would have produced, while leaving the visible text unchanged, makes the answer track the never-written value in 0.74 to 1.00 of cases across six models from four families (Qwen, Phi, OLMo, DeepSeek-R1-Distill) on arithmetic and narrative state-tracking tasks; a size-matched random perturbation moves the answer in at most 0.01 for instruction-tuned models. Blocking attention from later positions to that single token drops the answer's dependence on the value to at most 0.007 on four models, while blocking a neighboring token, a random prompt token, or an earlier written value changes nothing, and the model keeps producing fluent parseable answers. Two values planted at two positions compose into their sum, a number appearing nowhere in the input. The workspace does not substitute for this memory: with writing forbidden, the same models fail beyond one or two dependent steps. Models read the written value by default even when the prompt implies a different one; which evidence prompts recomputation instead is model-specific. For oversight this is direct: on serially dependent state a model cannot cheaply regenerate, reading the chain of thought observes the memory the computation actually uses.

1 Introduction

A fitted Jacobian lens reads a model's currently active concepts out of its residual stream, and this J-space workspace is a natural candidate for where multi-step reasoning happens [Gurnee et al., 2026]: on that picture the chain of thought reports work done internally, and interventions on the workspace should matter more than interventions on the text. We tested both locations with causal interventions and found the reverse: the serial state of the computation lives in the written trace, held at the written tokens as memory the model reads back, and the internal workspace neither holds a chain of dependent values nor carries the chain state in its lens-readable concept subspace (J-space). Concretely:

1. **The state at a written value's token is a value register.** Planting a never-written value into the residual stream at that token, text unchanged, makes the answer follow the planted value at 0.74 to 1.00 across six models in four families; restoring the correct state under corrupted text reverts the answer at 0.88 to 1.00; size-matched random noise does at most 0.01 on instruction-tuned models (Section 4).
2. **The register is read through attention to its own token.** Cutting attention from later positions to that token collapses the answer's dependence on the value to at most 0.007 on four models in three families, with neighbor-token, random-token, and earlier-value cuts all at the unblocked level, and with unparseable-output rates unchanged, so the cut removes the value rather than the ability to generate (Section 5).
3. **Registers compose.** Two never-written values planted at two positions yield their sum, which exists nowhere in the input, at 0.88 to 0.90; answers using only one of the two planted values occur at 0.01 (Section 4).
4. **The internal workspace does not substitute.** With writing forbidden, models from 4B to 671B fail beyond one to two dependent steps. Ablating the top-k active J-space concepts at a calibrated, generation-preserving dose changes neither accuracy nor the amount written, on synthetic chains and on GSM8K, so the chain state is not in the lens-readable subspace at that dose (Section 3).
5. **Reading is the default; checking is model-specific.** Models follow an edited written value even when the prompt implies the correct one. DeepSeek V3.2 recomputes only values the prompt generatively defines (reversion 0.43, targeted precisely at the defined value); Claude Sonnet 4.5 also uses stated checkpoints; Llama 3.3 70B checks almost nothing despite solving the same chains from scratch (Section 6).

Causal interventions on written reasoning tokens exist in prior work: Shih et al. [2026] patch scratchpad representations on a fine-tuned model (following 0.80 to 0.91) and Zhu et al. [2025] intervene on value-storing tokens in multiplication and dynamic programming. Our contribution is the generalization to unmodified pretrained models across four families and two task formats, the attention-level localization with matched controls and a fluency check, the composition result, the J-space boundary, and the map of when models check the record. Together these support a specific picture: the chain of thought functions as addressable memory, written once and read on demand, and the visible trace is causally load-bearing exactly where the model cannot cheaply regenerate the state it stores.

2 Tasks and protocol

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events, with varied names, items, verbs, and sentence forms. In both, the model writes one line per step giving the running value.

The protocol takes a correct worked solution, changes the value at one step j, truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, which is 0.00 throughout. Generation uses temperature 0.6 and top-p 0.95 with five samples per item; answers are parsed from a fixed final-line format and unparseable outputs are counted separately. Reported intervals are 95 percent bootstrap intervals resampling items. We report intervals rather than significance tests; the effects that carry the argument exceed their controls by margins far larger than the intervals. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill. Model revisions, prompts, layer bands, and per-run configurations are in the appendix; code and data will be released.

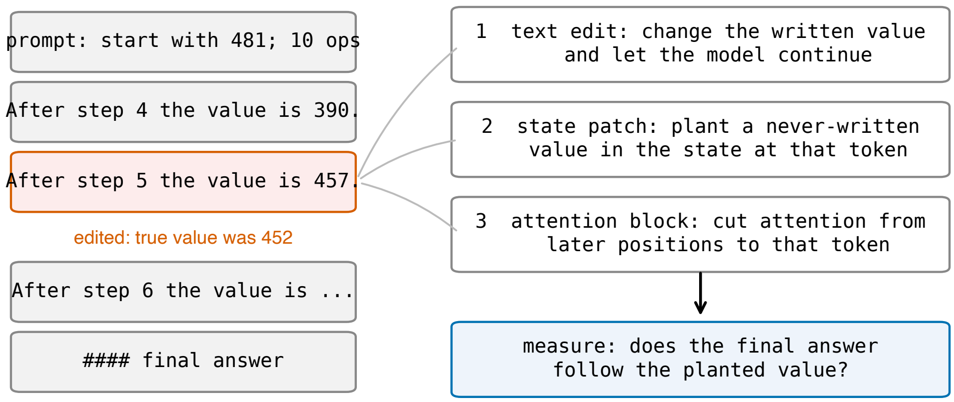

Figure 1: The edit-and-continue protocol. A correct worked solution is truncated after one edited value line and the model continues. Three interventions target the edited value: changing the visible token, overwriting the internal state at that token with the state a different value would have produced, and blocking attention from later positions to that token. The readout is whether the final answer follows the planted value.

3 The internal workspace does not hold the chain

Two results bound where reasoning state is not. First, the workspace cannot replace writing. Instructed to output only the final answer, with token counts confirming nothing else was generated, models fail beyond one to two dependent steps: Qwen2.5-7B falls from 1.00 at $d = 1$ to 0.04 at $d = 4$; DeepSeek V3.2 (671B) reaches 0.23 at $d = 1$ and near zero above. Models also write from the start rather than upon overflow: the fraction of ground-truth intermediates appearing in free traces is 0.93 to 1.00 at every difficulty from one step to 64, for every model from 1.5B to 671B.

Second, the chain state is not carried in J-space, the concept subspace read out by a fitted Jacobian lens (a per-layer map whose readout is the concepts currently active; ablation projects out the preimages of the top-k active concepts). At the strongest dose that leaves ordinary text generation intact (calibrated and frozen on neutral text before any task run: next-token agreement 0.87, perplexity ratio 1.12), removing the top 16 active concepts at every position changes neither chain-of-thought accuracy nor bare-answer accuracy nor the amount written, on synthetic chains and on GSM8K, and does no more than a size-matched random-directions ablation. The dose-response above this threshold is unexplored, so the result bounds the lens-readable subspace at generation-preserving doses rather than ruling out any internal carrier.

4 The written value is an addressable register

Table 1 decomposes the effect of editing a written value. Restoring the correct internal state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting a third value that appears nowhere in the text makes the answer track it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. Because different planted values produce answers following those values, the value content is the causal quantity, and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists with depth: on Qwen3-4B the planted value is followed at 0.98, 0.97, and 0.95 for chains of 5, 20, and 40 steps.

The result is a property of written state, not of one template. On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

Table 1: State patch on arithmetic chains at ten steps. Rates are conditioned on items where the text edit was followed (first column), so cross-model comparison of later columns should account for that rate. Brackets are 95 percent bootstrap intervals. The two reasoning-distilled models re-solve the problem after their reasoning phase, which raises their random-control reversion; in a logged sample every reverting continuation re-derives from the problem's starting value rather than correcting in place, and re-derivation can only produce the correct answer, so the planted-value column is unaffected.

model	text edit followed	restore correct	plant new value	random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B-Instruct	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

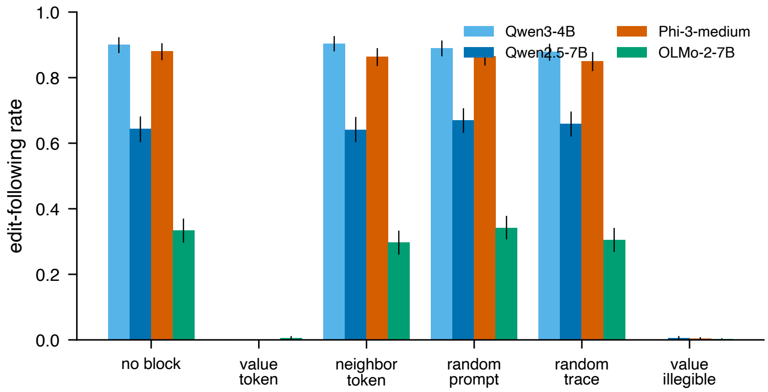

Figure 2: Rate at which the answer follows the edited value under each blocked attention target. Blocking the value's own token collapses following on every model; matched control cuts do not. Error bars are 95 percent bootstrap intervals.

Registers compose. In a task with two counters summed at the end, planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The model reports a number assembled from two independently set registers.

5 The register is read through attention to its token

Blocking attention from all post-edit positions to the edited value's token collapses the answer's dependence on that value on every model tested (Table 2, Figure 2). Matched cuts to the adjacent words, to a random prompt token, or to an earlier step's written value leave the dependence at its unblocked level. Cutting the operand the next step needs breaks the arithmetic itself, yielding answers that match neither the edit nor the correct value. The collapse holds across three families and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on Phi-3-medium).

The cut removes the value, not the ability to generate. Under the value-token block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number that matches neither the edited value nor the correct one: it computes forward from a value it can no longer retrieve. The intervention removes access to one value rather than disabling generation near the position.

Table 2: Attention block on arithmetic chains: rate at which the answer follows the edited value. OLMo-2-7B continues this format less reliably (low no-block rate) and shows the same contrast.

model	no block	value token	neighbor	random token	earlier value
Qwen3-4B	0.900	0.000	0.903	0.778	—
Qwen2.5-7B-Instruct	0.643	0.000	0.640	0.670	0.658
Phi-3-medium	0.880	0.000	0.863	0.865	0.850
OLMo-2-7B	0.333	0.005	0.297	0.342	0.305

Reading is the preferred path, not the only one. Replacing the value's characters with unreadable dots, without touching attention, also removes following, and differently by model: the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases, while Phi-3-medium does not recover (0.00). When the token is legible the model reads it; when it is not, some models rebuild it.

Localization within the network. The block masks the token at every layer and the patch overwrites a band of middle layers, so these results establish that the value is load-bearing and that attention to its token is the conduit, without yet fixing the depth at which the read occurs. A layer-resolved sweep of both interventions is in progress; preliminary runs indicate the patch is sufficient at early-to-middle layers and the cut is necessary at middle layers, and the full sweep will be reported in the final version.

6 When the written value is checked

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted to correct at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the check targets exactly the defined value. Claude Sonnet 4.5 checks more broadly, reverting most when a stated checkpoint is the value's only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so capacity to recompute does not imply checking. Qwen models at 4B and 7B never revert. These conditions come from separate experiments with different trace lengths and item sets; each within-experiment contrast is controlled, and we do not claim a single ordered scale across models or a monotonic trend in capability.

The same test bounds the mechanism on natural benchmarks: on GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, because a benchmark intermediate is often one operation from numbers still in the prompt and whether a model rereads or recomputes depends on its own capacity. Written values act as memory precisely for state that the model cannot cheaply reconstruct.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which motivates the layer sweep in progress. The closest work intervenes on written reasoning tokens: Shih et al. [2026] on a fine-tuned model, Zhu et al. [2025] on multiplication and dynamic-programming tasks. We add unmodified pretrained models across four families and two formats, attention-level localization with matched controls and a fluency check, composition, the J-space boundary, and the checking map. Faithfulness studies show stated reasoning and internal computation can diverge [Turpin et al., 2023, Lanham et al., 2023]; filler-token results show computation can pass through uninformative tokens [Pfau et al., 2024]; both bound our claim to state the model cannot regenerate. Expressivity theory predicts long dependency chains must live in generated text [Merrill and Sabharwal, 2024]; Self-Notes [Lanchantin et al., 2023] studies writing as memory behaviorally.

8 Limitations

The patch overwrites a band of middle layers and the block masks all layers; the layer-resolved localization is in progress and the current results do not yet fix the depth of the read. The J-space result bounds only generation-preserving doses of one fitted lens. White-box evidence is at 4B to 14B; frontier evidence is behavioral. Tasks are arithmetic and narrative-count chains; richer state such as code or multi-entity plans is untested. The checking-policy results support within-experiment contrasts, and the only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

The chain of thought is where a model's serial reasoning state lives. Written values are held at their tokens as addressable registers, read back through attention, composable, and decisive for the answer even against the text and against the prompt; the internal workspace neither holds chains of dependent values nor carries them in its lens-readable J-space subspace at doses that preserve generation. Reading the trace therefore observes the memory the computation uses, on exactly the state the model cannot cheaply regenerate, and diverges from the computation where it can.

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